

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics and Economics
Honour School of Economics and Management
Honour School of History and Economics
Honour School of Engineering, Economics and Management (Part II)
Honour School of Materials, Economics and Management (Part I)

MICROECONOMICS

TRINITY TERM 2017

Friday 26th May 2017, 14:30 - 17:30

Please start the answer to each question on a separate page.

There are 10 questions in this paper.

*Answer **all** questions in Part A and **two** long questions from Part B.*

Part A attracts 1/3 of the marks in total; each question in Part B attracts 1/3 of the marks.

In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets;

in Part B, the weight assigned to each part of the multipart question is indicated in square brackets.

Candidates may use their own calculators.

Do not turn over until told that you may do so.

Part A

In Part A, the weight assigned to each question (as a percentage of the total for Part A) is indicated in square brackets

1. Consider an economy which produces output X using labour input L according to the production function $X = 2L^{1/2}$, and which contains one consumer with the utility function $u = X - \frac{1}{2}L^2$.

- (a) If the consumer controls production directly, what quantity of labour input will the consumer choose?
- (b) Define a general competitive equilibrium.
- (c) Suppose now that the price of X is p and the price of L is w .

Production is chosen by a firm with the production function given above to maximize profits. Derive the firm's labour demand function and its level of profits as functions of p and w .

The household has the utility function given above, which it maximizes subject to its budget constraint. Derive the household's labour supply function.

What is the equilibrium price-wage ratio of this economy, and what is the equilibrium level of employment and output?

Explain your result in terms of the first theorem of welfare economics.

Outline briefly how your answer would change if there were a tax on wage income in this economy.

[30%]

2. Two firms sell identical products and are the only firms in the industry. They have the same constant and positive marginal cost, no fixed costs, and they set prices non-cooperatively.

Explain why, in theory, a merger between the duopolists creating a profit-maximizing monopolist could increase social welfare.

Could such a merger also benefit consumers?

(A well-annotated diagram could form the basis of your answer.)

[15%]

3. Janet's utility as a function of wealth is $u(w) = 2w^{1/2}$.

- (a) Show that her coefficient of absolute risk aversion is decreasing in initial wealth, but her coefficient of relative risk aversion is constant.
- (b) She has initial wealth $w = 104$ and is facing the lottery $(\frac{1}{2}, -40; \frac{1}{2}, +40)$.
What is her certainty equivalent of this lottery, and what is the associated risk premium?
- (c) Sam has the same utility function and initial wealth, but is instead facing the lottery $(\frac{1}{2}, -20; \frac{1}{2}, +20)$.
What, *approximately*, is the ratio of Sam's risk premium to that of Janet from the previous part, and why?
(You should not recalculate the certainty equivalent for Sam's lottery.)

[25%]

4. There are two types of worker: H -types with high productivity θ_H , and L -types with low productivity θ_L ; each worker knows their own type. Many risk-neutral firms compete for the services of the workers, but they cannot observe a worker's type.

- (a) Why would some of the workers like to be able to signal their type to the firms?
What condition(s) would make a signal credible?

Suppose that $\theta_H = 240$, $\theta_L = 180$, and that there are half as many H -types as L -types; also, the workers have no outside option.

- (b) When no signal is available, what is the market equilibrium outcome?

Suppose the workers can acquire an education, observable by the firms – this would cost only 45 for an H -type, but 55 for an L -type.

- (c) Consider an outcome in which H -types signal and L -types don't, the firms paying appropriate wages.
Is this outcome a market equilibrium? If not, what *is* the equilibrium outcome?
Is the equilibrium efficient?

[30%]

Part B

5. An economy experiences an increase in the cost of exporting (equivalent to a fall in the price it receives for its exports).
What is the general equilibrium effect of this on the economy?
How does it change the structure of production, the returns to factors of production, and household welfare?

How is your answer modified if the economy also experiences a reduction in labour supply?

Do your answers provide insight into the possible economic impact of Brexit?
6. What criterion should society use to compare different social outcomes?
Can such a criterion be derived from individuals' preferences?
Can it be used in practice to evaluate the value of policy changes?
7. Does the analysis of collusion using the theory of repeated games give useful hints to oligopolists who are trying to collude?
Does the analysis help anti-trust authorities who would like to deter collusion?
8. Under what conditions is expected utility theory a useful guide to decision making in the face of uncertainty?
9. Hugh believes that he is low risk – he locks windows and doors when he leaves home, and he doesn't leave candles burning when he goes to bed. He faces insurance that is actuarially unfavourable, and so decides not to buy any. Is he acting optimally?

How might competitive insurance companies overcome the problem of serving both low risk and high risk individuals?

10. A risk-neutral farm owner (the principal) hires a risk-averse farm worker (the agent) to manage this year's crops. The harvest will be either small, worth only 10 to the farm owner, or large, worth 50. The size of the harvest depends on which of three effort levels that the worker exerts, as summarised in the table.

	small (10)	large (50)
Pr e_H	1/5	4/5
Pr e_M	4/5	1/5
Pr e_L	1	0

The worker's utility from receiving a wage w and exerting effort e is $\sqrt{w} - e$, where $e = 3$ if effort is e_H , 2 if effort is e_M , and 1 if effort is e_L . The worker has a reservation utility of 0.

- (a) When both effort and the size of the harvest are observable, what does the owner consider when deciding which contract to offer the worker?

What contract will the owner choose? [20%]

Now assume that effort is unobservable. The wage in the contract between the two parties will depend only on the size of the harvest: w_1 if it is a small harvest, w_2 if it is large.

- (b) Write v_i for the value to the worker of a wage w_i , i.e. $v_i = \sqrt{w_i}$.

If the worker would prefer to exert effort e_M rather than e_H , what is the largest value that $v_2 - v_1$ can be?

If the worker would prefer to exert effort e_M rather than e_L , what is the smallest value that $v_2 - v_1$ can be?

Hence, or otherwise, show that there is no contract under which it is incentive compatible for the worker to choose e_M . [30%]

- (c) Explain briefly, and in general, (i) how the owner would implement low effort, e_L , (ii) how he would implement high effort, e_H .

What is the optimal contract in this case? [35%]

- (d) Why does the owner suffer from effort being unobservable, and by how much? [15%]